Assignment 1 - Git & GitHub Fundamentals Report

- Full Name: Rayan Benlachhab
- Student ID: 158274
- Course Code & Section: CSC332 01
- GitHub Username: benlachhabrayan-github

1. Repository Setup Briefly explain how you initialized the repository and what you ignored in .gitignore.

For initializing the repository, I just entered on the git bash “git init” and it was prepared.
On the “.gitignore” file, I ignored the “local.env” and all files that had “.log” at the end of their names letting only “important.log”, the only exception not affected to be shown.

2. README & Commit History Briefly describe your README and how you organized your commits.

On my “README.md” file, I wrote my name, my ID GitHub, the technology I used for the homework, a piece of code in Python, a Youtube link, and a picture that was already included on the initial starter file in the “assets” file.
For the commits, I did 6 in total: the first was right after initializing the README.md and .gitignore files, the second was an update I made on the notes.txt, the third was after making updates on two files but only adding one to be commited, the fourth was when I added the about section from the feature/about branch, the fifth was a simple update I made in “style.css” file, and the last was when I updated from Github a change I made on the README.md file

```
PROBOOK@DESKTOP-7L7SVS9 UCRT64 ~/Downloads/git-session1-benlachhabrayan-glitch (main)
$ git log --oneline --graph --all
* 1ddceff (HEAD -> main, origin/main, origin/HEAD) Add line from GitHub to README
* e77cbc4 update changes in style.css
* 2465828 (feature/about) added about section
* 21480c6 update notes.txt and not update app.js even though I modify it
* efb273a add notes.txt
* 6b5556f add starter files
```

3. Selective Staging Explain how you changed two files but committed only one of them first.

I added new lines to both “app.js” and “notes.txt” files updating them. However, I added only notes.txt using git add. As a result, when I committed the changes, only notes.txt was included in the commit, while the changes to app.js remained in my local branch.

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.js
```

4. Branch & Merge Briefly explain the feature branch you created and how you merged it into main.

I started by switching from the main branch to a new created one called “about” using the switch -c feature/about. When I did added lines on the “README.md” file and committed it, I returned back to the main branch by using "switch main" and I merged the two branches using "git merge feature/about" that made the update I commited on the “about” branch to merge into my main branch

```
PROBOOK@DESKTOP-7L7SVS9 UCRT64 ~/Downloads/git-session1-benlachhabrayan-glitch (main)
$ git branch
  feature/about
* main
```

5. Remote Changes Explain the difference between commit and push, then fetch and pull.

The difference between commit and push is that committing save changes on the local repository, where push saves the local repository changes on the remote repository on github.
The difference between fetch and pull is that fetching takes the update from the remote repository but doesn't merge them with your local one, and pulling is taking the updates fetched and merge them with your local repository

6. Fair Use of AI

Yes I used AI, I used it to help me guide through the homework because it was my first experience with Gi. I was asking constantly for questions related to each part of the assignment, either to show me what to do or to explain me the purpose of certain codes. He was always right, he didn't gave me any wrong answers and it always matched the results that I was finding in git bash.
The lab was essential for helping me understand the concepts, while AI helped me better understand how to apply them in practice.

7. Sources & Reflection List any useful sources you used. Then write 2-3 sentences on the hardest concept and what you would check first if Git behaves unexpectedly

The only sources I used through the homework was ChatGpt and a bit of my own knowledge that I gained through the LAB and Github's documentation.
The hardest concept was to understand how branches work, even though it was explained in the LAB, I believe that trying to understand by yourself with practices is more efficient than just rely entirely on theories.

If Git ever behave unexpectedly, I would firstly look at what would be written for the problem on the Git bash, if I can recognize it and solve it, it would be fine, but if no, I will ask directly an AI to answer me because it has a large database to understand which problem I’m with quickly.